

Class 7 Biology — lesson plan

- Period 3, 40 minutes. Bring a potted plant and a leaf for the class to look at.
- Attendance first. Collect the homework notebooks from last week.
- Remind the class about the field trip form due on Friday.

1. Photosynthesis — meaning

- Photosynthesis is the process by which green plants prepare their own food using sunlight.
- Break the word: photo = light, synthesis = putting together.
- Plants are autotrophs (producers); animals depend on them for food.

2. What the plant needs

- Four raw materials: sunlight, water, carbon dioxide and chlorophyll.
- Roots absorb water from the soil; xylem carries it up to the leaves.
- Carbon dioxide enters through stomata, the tiny pores on the underside of the leaf.
- Chlorophyll, the green pigment in chloroplasts, traps the light energy.

3. Steps of the process

- Step 1: chlorophyll absorbs light energy.
- Step 2: light energy splits water molecules into hydrogen and oxygen.
- Step 3: hydrogen combines with carbon dioxide to form glucose.
- Step 4: oxygen is given out as a by-product through the stomata.

4. Word equation

- Carbon dioxide + water --(sunlight, chlorophyll)--> glucose + oxygen.
- $6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$ (only for the faster students).
- Glucose is stored as starch; iodine test turns it blue-black.

5. Why it matters

- Source of food for almost every living thing on Earth.
- Releases the oxygen we breathe; removes carbon dioxide from the air.
- Fossil fuels such as coal came from plants that stored the sun's energy long ago.

6. Respiration (compare next class)

- Respiration breaks down glucose to release energy; it happens day and night in all living cells.
- Glucose + oxygen $\rightarrow$ carbon dioxide + water + energy: the reverse of photosynthesis.
- Homework: table comparing photosynthesis and respiration.